Literature Review

Gohr (2019)

Gohr (2019) provides the foundational study for the present research because it demonstrates that deep neural networks can contribute directly to the cryptanalysis of a modern reduced-round block cipher. The central problem addressed is whether a neural network can identify statistical properties in ciphertext pairs that are not fully represented by conventional differential cryptanalysis. The study focuses on SPECK32/64 and assumes a chosen-plaintext setting in which the attacker can generate plaintext pairs with a selected input difference. It also assumes that reducing the number of encryption rounds leaves sufficient non-random structure for statistical learning to remain possible.

The methodology combines classical and neural cryptanalysis. Gohr (2019) first computes the complete difference distribution associated with the input difference `0x0040/0000` for up to eight rounds under a Markov model. This provides a strong classical baseline rather than comparing the neural network only with random guessing. A residual neural network is then trained on labelled ciphertext pairs, where positive samples originate from plaintext pairs with the selected difference and negative samples are generated from random pairs. The neural output is subsequently used to rank final-round subkey guesses after partial decryption. A selective Bayesian key-search procedure is also introduced to avoid evaluating every candidate with equal effort.

The results show that the neural distinguishers achieve stronger classification performance than the complete classical difference-distribution model in the evaluated setting. In a nine-round partial-key experiment using 128 chosen plaintexts, the neural method produces a mean correct-key rank approximately five times lower than the classical comparison. The complete framework also supports an estimated attack on 11-round SPECK32/64 with a computational complexity of approximately (2^{38}) reduced cipher operations, compared with approximately (2^{46}) for the cited classical attack.

However, the results apply only to reduced-round SPECK32/64 and do not constitute an attack on the complete 22-round cipher. Training requires large synthetic datasets, neural inference is slower than a table lookup, and the learned features are only partly understood. The model also produces probabilistic rankings rather than exact key verification. For the present project, Gohr's work supplies the main neural-distinguisher architecture and subkey-ranking baseline. The proposed research extends this framework by passing ranked subkey candidates to an exact SAT solver and evaluating whether the neural ranking reduces total master-key recovery cost.

Chen, Shen, Yu, and Yuan (2023)

Chen, Shen, Yu, and Yuan (2023) examine whether neural distinguishers can be strengthened by processing multiple ciphertext pairs simultaneously rather than classifying each pair independently. Their research problem is based on the observation that conventional Gohr-style distinguishers learn features contained within one ciphertext pair, even though several pairs may collectively expose stronger distributional information. The authors aim to determine whether this aggregate information improves distinguishing accuracy and whether the improvement remains useful when the distinguisher is incorporated into key-recovery attacks.

The study assumes that groups of ciphertext pairs contain derived distributional features that are difficult to infer from a single sample. An important part of the investigation concerns whether the ciphertext pairs must share the same encryption key. Earlier experiments achieved very high accuracy by grouping ciphertexts generated under one key, creating the possibility that the model learned key-dependent behaviour rather than a generally useful cipher distribution. Chen, Shen, Yu, and Yuan (2023) therefore compare groups generated under one key with groups in which individual ciphertext pairs use independently selected keys.

The methodology evaluates single-pair and multiple-pair neural distinguishers on SPECK32/64, Chaskey, PRESENT, DES, and SHA3-256. The authors construct neural inputs containing several ciphertext pairs and introduce false-negative and false-positive tests to investigate whether the model has learned features arising from the combined distribution. They also adapt the new distinguisher to Gohr's key-ranking attack and the Neural-Aided Statistical Attack. A data-reuse strategy is proposed to reduce the amount of newly generated attack data.

The results show that processing multiple ciphertext pairs generally improves distinguishing accuracy across the evaluated primitives. The improvement remains when different keys are used within one input group, indicating that shared-key information is not the primary explanation. The stronger distinguisher also reduces the data complexity of the statistical attack. However, it does not produce a significant improvement in Gohr's original key-recovery framework because that attack is not determined solely by classifier accuracy.

The main limitation is that larger groups increase input size, memory consumption, and inference cost. The study also remains limited to reduced-round experiments and probabilistic recovery methods. For the present project, the work supports aggregating several ciphertext pairs when scoring a subkey and motivates confidence measurements based on ranking stability across independent pair subsets.

Lu et al. (2022)

Lu et al. (2022) extend differential neural cryptanalysis beyond SPECK by developing improved ordinary and related-key distinguishers for the SIMON and SIMECK cipher families. Their research problem is that neural-distinguisher performance depends not only on the selected network architecture but also on the cryptographic information supplied to the model. A neural network trained directly on raw ciphertext pairs may be required to rediscover relationships that are already known from the cipher structure. The authors therefore investigate whether domain-specific features, multiple ciphertext pairs, and carefully selected input differences can improve distinguishing performance.

The underlying assumption is that cryptographic feature engineering can provide an inductive bias that helps the neural model isolate relevant differential behaviour. Each training sample contains eight ciphertext pairs together with ciphertext differences and approximations of differences from preceding rounds. The selected architecture is a squeeze-and-excitation residual network, which assigns different importance to individual feature channels. The study also evaluates how the choice of the neural-distinguisher input difference and the connecting difference between the classical and neural components affects hybrid distinguishing performance.

Experiments are conducted on SIMON32/64, SIMON64/128, SIMECK32/64, and SIMECK64/128. Both ordinary-key and related-key attack settings are considered. Lu et al. (2022) improve several previously reported neural-distinguisher accuracies and present related-key neural distinguishers for all four evaluated cipher variants. For example, the reported accuracy for nine-round SIMON32/64 is approximately 0.918 using eight ciphertext pairs. However, accuracy declines sharply as the number of rounds increases and approaches random classification at the highest evaluated round counts.

The study demonstrates that multiple pairs and cipher-specific representations can improve neural performance, but it also confirms that feature engineering does not remove the higher-round loss of signal. The related-key setting additionally grants the attacker stronger capabilities than the ordinary chosen-plaintext model. The experiments were conducted using server-class processors, 512 GB of memory, and an NVIDIA T4 GPU, reducing the direct reproducibility of the complete setup on personal hardware.

For the present research, Lu et al. (2022) provide evidence that structured cryptographic features may improve subkey ranking but should not be trusted equally across all round configurations. This finding supports a confidence-gated system that uses neural guidance only when its ranking signal is sufficiently strong.

Bellini et al. (2023)

Bellini et al. (2023) address the lack of generality in neural cryptanalysis. Following Gohr's original study, many researchers developed highly specialised architectures, input representations, and training procedures for individual ciphers. Although this produced strong results for particular targets, it required substantial manual cryptanalytic and machine-learning effort before the method could be applied to another primitive. Bellini et al. aim to construct a cipher-agnostic pipeline that automatically selects promising input differences and trains a neural distinguisher without extensive cipher-specific architecture tuning.

The proposed pipeline contains two principal components. The first is an evolutionary optimisation algorithm that searches the space of possible input differences. Candidate differences are evaluated using an inexpensive neural distinguishing score, allowing the search to avoid exhaustive enumeration as the block size increases. The second component is DBitNet, a generic bit-oriented neural architecture based on dilated convolution. The same basic architecture and training procedure can be used across ciphers with different round structures, state sizes, and nonlinear operations.

The pipeline is evaluated on SPECK, SIMON, PRESENT, KATAN, TEA, XTEA, LEA, HIGHT, and GIMLI. These targets cover ARX constructions, Feistel networks, substitution–permutation networks, and cryptographic permutations. Bellini et al. (2023) show that the automated framework can match or exceed specialised approaches in several comparable settings. It reproduces an eight-round SPECK32 distinguisher with performance similar to the specialised baseline and produces new distinguishers for primitives that had not previously received equivalent neural analysis.

The main limitation is that the framework is designed to discover basic neural distinguishers rather than complete key-recovery attacks. It does not automatically integrate prepended differential characteristics, neutral bits, partial decryption, key ranking, or exact key verification. The authors identify related-key analysis, combinations of several input differences, integration with classical attacks, and the construction of complete key-recovery procedures as future directions. Training models for numerous differences and round configurations can also require substantial computing time.

For the present project, the work provides a valuable reproducibility and automation baseline. However, automated difference selection should remain an optional experiment rather than the main contribution. The stronger project gap concerns determining when neural ranking should be trusted during exact key recovery and when the solver should ignore it.

Baksi et al. (2023)

Baksi et al. (2023) investigate whether useful machine-learning distinguishers require highly specialised deep-learning architectures. Their research problem is that much of the

neural-cryptanalysis literature assumes an attacker has sufficient expertise and computing resources to construct large, cipher-specific convolutional networks. The authors instead evaluate whether comparatively simple multilayer perceptrons and support-vector machines can still detect non-random behaviour in reduced-round cryptographic primitives.

The study assumes that classical differential distinguishers may underestimate an attacker by discarding output differences that do not match one selected characteristic. A machine-learning classifier can process a broader output representation and potentially combine several weak statistical signals. The experiments use reduced-round and predominantly unkeyed versions of SPECK-32, SPECK-128, ASCON, SIMECK-32, SIMECK-64, and SKINNY-128. Different input differences, activation functions, neural-network libraries, and binary or multiclass classification settings are evaluated. Most experiments are deliberately performed using personal-computer-scale hardware.

The results include distinguishers for eight-round SPECK-32, seven-round SPECK-128, three-round ASCON, nine-round SIMECK-32, fourteen-round SIMECK-64, and selected SKINNY-128 configurations. The eight-round SPECK result reaches the same round coverage as more specialised neural architectures while using simpler models and comparatively low training complexity. The authors also report different results between TensorFlow/Keras and PyTorch under similar settings. This is particularly important because distinguishing accuracies of approximately 0.51 may be treated as meaningful in cryptanalysis even though such small differences can be affected by implementation details.

Several limitations reduce the direct security implications of the results. Many experiments use unkeyed permutations or idealised independent round keys rather than the complete cipher key schedule. Some hyperparameters and input differences are selected experimentally without a unified optimisation procedure. Marginal performance above random classification may also be caused by dataset artefacts, framework behaviour, or random variation unless it is reproduced across independent datasets and seeds. Most importantly, the paper does not implement complete key recovery.

For the proposed project, Baksi et al. (2023) provide appropriate lightweight baselines and reinforce the need for repeated trials, confidence intervals, correct-key ranking, and end-to-end attack measurements rather than reporting classification accuracy alone.

Liu et al. (2023)

Liu et al. (2023) focus on the computational cost of neural differential cryptanalysis. Their research problem is that attempts to improve neural-distinguisher accuracy commonly increase the number of input features, network layers, trainable parameters, and required training examples. This may make neural cryptanalysis expensive and reduce its applicability on devices

or research systems with limited computing resources. The authors aim to reduce training cost through both a smaller data representation and a more efficient neural-network architecture.

The first part of the proposed method introduces an advantage-bit search procedure. Instead of supplying the complete ciphertext pair to the model, the procedure identifies a small subset of output-difference bits expected to contribute strongly to distinguishing. Two cryptographic conditions guide the search. A symmetry condition reflects relationships between the left and right words of SPECK, while a difference condition allows the complete ciphertext representation to be replaced by selected difference bits. The second part of the method replaces conventional convolutions with depthwise separable convolutions, reducing the number of parameters and arithmetic operations.

Experiments on reduced-round SPECK compare full-input Gohr-style models, partial-output-difference distinguishers, and architectures using separable convolution. Liu et al. (2023) report that selected four-bit and six-bit input representations can retain meaningful distinguishing performance while greatly reducing the amount of input information. The separable-convolution architecture reduces the parameter count while producing only a limited accuracy reduction in selected configurations. When the reduced representation and the lightweight architecture are combined, the resulting model uses approximately 30% of the parameters associated with the referenced Gohr model.

The principal limitation is that parameter count is treated as the main measure of efficiency. A smaller model does not necessarily produce a faster complete attack because candidate generation, repeated partial decryption, neural evaluation over many subkeys, and key verification may dominate the total cost. The evaluation is also concentrated on SPECK despite broader claims concerning generality. Alternative efficient architectures and wider cipher evaluation remain open.

This study is relevant to the proposed MacBook-based implementation because it shows that useful neural distinguishers can be constructed with reduced computational requirements. Model compression should nevertheless remain an implementation ablation rather than the central research contribution.

Chen, Shen, and Yu (2023)

Chen, Shen, and Yu (2023) introduce the Neural-Aided Statistical Attack to address limitations in earlier neural key-recovery methods. Gohr's attack relies partly on experimentally selected neural-score thresholds, making it difficult to estimate the required data complexity before performing extensive experiments. It also commonly depends on neutral bits in a prepended differential characteristic to construct sufficiently large plaintext structures. The authors aim to

develop a more general neural-aided attack that supports theoretical complexity estimation and remains usable when suitable neutral bits are unavailable.

The proposed method treats each thresholded neural-distinguisher output as a Bernoulli observation. For a collection of ciphertext pairs, the number of neural outputs exceeding a selected threshold becomes a statistical variable. The right-key and wrong-key cases are approximated using normal distributions, allowing the required number of samples and the decision threshold to be estimated from predefined error probabilities. Key recovery is therefore converted into a statistical hypothesis test rather than relying only on an unanalysed neural score.

Three optimisation methods are introduced. First, the Bit Sensitivity Test identifies informative ciphertext bits, enabling the attacker to train a distinguisher using only bits influenced by a smaller subset of subkey variables. Second, a Bayesian key-search strategy prioritises promising candidate values rather than evaluating the complete candidate space uniformly. Third, neutral bits can still be used when available to reduce the required data complexity, although they are no longer a mandatory condition. Experiments are conducted on reduced-round SPECK32/64, DES, and SPECK96/96.

Chen, Shen, and Yu (2023) demonstrate nine- and ten-round SPECK attacks without relying on neutral bits and provide theoretically motivated data-complexity estimates. Informative-bit selection reduces the number of key bits that must be guessed simultaneously, while the Bayesian strategy reduces unnecessary candidate evaluation. Nevertheless, the statistical model depends on approximate distributional assumptions and empirically observed neural behaviour. The final candidate list remains probabilistic and requires subsequent verification.

For the proposed research, the Neural-Aided Statistical Attack supplies an important comparison method. Neural–SAT recovery should be compared not only with random ordering and SAT alone but also with statistically defined candidate ranking. CryptoMiniSat can then provide exact consistency checking that the statistical attack itself does not offer.

Soos et al. (2009)

Soos et al. (2009) provide the principal SAT-solving foundation for the proposed research. Their central problem is that general-purpose SAT solvers typically accept Boolean formulas in conjunctive normal form, while cryptographic algorithms contain long XOR relations and systems of linear equations over (GF(2)). Directly expanding these relationships into ordinary CNF can create many clauses, increase memory consumption, and hide algebraic structure that could otherwise support more efficient reasoning.

The authors assume that cryptographic SAT performance can be improved only when the cipher representation and the solver implementation are adapted together. They extend MiniSat with

native XOR clauses so that one XOR equation can be retained internally rather than expanded into several ordinary clauses. Gaussian elimination is incorporated to reason across combinations of linear equations when standard DPLL propagation becomes ineffective. Visualisation tools are also developed to observe solver behaviour, identify bottlenecks, and guide modifications to cipher representations. Logical-circuit generation, Karnaugh-map simplification, and preprocessing are then applied to the encoded problems.

The modified solver is evaluated on three stream-cipher targets: Crypto-1, HiTag2, and Bivium. The results show that the value of each optimisation depends on the structure of the target. Crypto-1 recovery decreases from hundreds of seconds with a basic solver configuration to approximately 40 seconds with the combined improvements. The authors also report that Bivium is solved approximately 26 times faster than the previous SAT-based result. Native XOR support reduces formula size and memory consumption. However, Gaussian elimination is not universally beneficial and can reduce performance when the linear subsystem does not provide sufficient additional propagation.

The paper is limited by its target selection and age. It studies shift-register-based stream ciphers rather than an ARX block cipher containing repeated modular addition. The effectiveness of the same techniques on SPECK32/64 must therefore be tested rather than assumed. The work also does not include neural guidance.

Its relevance remains fundamental because it establishes that exact cryptanalysis depends strongly on preserving the cipher's algebraic structure. The proposed SPECK model should consequently use native XOR constraints, direct wiring for rotations, and exact carry constraints for modular addition.

Selsam and Bjørner (2019)

Selsam and Bjørner (2019) investigate how neural predictions can guide an exact SAT solver without attempting to replace it. Their research problem is that neural SAT models had demonstrated the ability to classify small formulas or identify unsatisfiable cores, but there was no established method for improving high-performance solvers on realistic benchmark instances. The authors propose using a neural model only periodically to redirect an existing branching heuristic.

The approach assumes that variables appearing in an unsatisfiable core are more likely to be useful branching choices. A simplified NeuroSAT graph network, called NeuroCore, is trained to predict the probability that each variable belongs to an unsatisfiable core. Generating sufficient labelled data is difficult because SAT competition collections contain relatively few suitable unsatisfiable formulas. The authors therefore recursively divide difficult formulas into smaller subproblems, run Z3 with a fixed timeout, extract DRAT proofs from solved unsatisfiable

instances, and identify variables that appear in the resulting cores. This procedure produces more than 150,000 labelled training examples.

NeuroCore is integrated into MiniSat, Glucose, and Z3 through periodic refocusing. Instead of querying the network for every branching decision, the solver periodically replaces its EVSIDS activity scores with values derived from the neural predictions. Local conflict-driven search then continues using the normal solver mechanisms. Selsam and Bjørner (2019) report that the modified solvers complete more SATCOMP-2018 instances within the selected timeout. A model specialised for scheduling instances produces greater improvement on its target distribution.

However, the approach has important limitations. Training-data generation requires expensive solver runs and proof extraction. Unsatisfiable-core membership is also a coarse supervision target because a core may contain many variables and not every core variable is an effective branching choice. Neural inference introduces substantial overhead, particularly when using a GPU. Later work further reported that some configurations did not clearly outperform random refocusing.

For the proposed project, the main contribution is the principle that learning should complement rather than replace the exact solver. The study also establishes the importance of including a random-guidance baseline and accounting for neural-inference overhead.

Han (2020)

Han (2020) revisits neural guidance for SAT solving and directly addresses several weaknesses identified in NeuroCore. The paper highlights three principal issues: dependence on GPU inference, the use of solver implementations that were no longer state of the art, and an expensive training pipeline based on unsatisfiable-core extraction. It also notes that later experiments suggested some NeuroCore improvements were not clearly better than random refocusing. The research objective is therefore to design a cheaper and more defensible neural target that can guide a modern SAT solver through CPU-based inference.

The selected supervision target is the glue variable. Glue clauses have a low literal block distance and are retained by modern conflict-driven solvers because they frequently contribute to useful propagation. Han modifies CaDiCaL to run for a bounded period and record how often each variable appears within a glue clause. These counts are converted into a probability distribution and used to train a compact graph neural network. Unlike unsatisfiable-core extraction, glue-variable labels can be collected before a solver completes and from the normal search process.

The trained model periodically refocuses CaDiCaL's variable scores. Its architecture is designed to support CPU inference even for formulas containing large numbers of clauses. Supervised models are evaluated on SATCOMP-2018 and SATRACE-2019. Han (2020) also formulates glue-variable prediction as a reinforcement-learning task and trains a distribution-specific model using SAT encodings of SHA-1 preimage attacks. The reported experiments show improvements over both the unmodified solver and a random-refocusing baseline in selected settings.

The work remains a preprint, and the observed benefit depends on the target distribution, refocusing interval, model architecture, and solver configuration. Modifying CaDiCaL's internal branching strategy would also create considerable implementation risk for a short university project. SHA-1 preimage instances differ structurally from SPECK key-recovery formulas, so the reported cryptographic result cannot be transferred directly.

For the present research, Han (2020) provides a strong methodological precedent for lightweight, distribution-specific neural guidance. The proposed project can apply the same principle more conservatively by ranking a 16-bit SPECK subkey space once and using incremental SAT assumptions instead of modifying every internal solver decision.

Zheng et al. (2024)

Zheng et al. (2024) investigate the representation problem in neural SAT cryptanalysis. Existing graph-based neural SAT solvers commonly encode formulas in conjunctive normal form using a bipartite clause–literal graph. This representation is suitable for conventional SAT problems but can be inefficient for cryptographic formulas because one XOR relationship may expand into many ordinary clauses and auxiliary variables. The authors aim to preserve cryptographic operations more directly through algebraic normal form.

The study assumes that ANF provides a more concise and semantically meaningful representation of XOR, AND, and higher-order Boolean relationships. Cryptographic operations including rotations, bitwise XOR, bitwise AND, and modular addition are converted into systems of equations over (GF(2)). Clauses, variables, and higher-order monomials are represented as structured graph components. CryptoANFNet, a message-passing graph neural network, is then trained to predict whether a plaintext–ciphertext cryptographic instance is satisfiable.

The evaluation compares CryptoANFNet with NeuroSAT and heuristic solvers on small- and larger-scale datasets derived from encryption algorithms. Zheng et al. (2024) report classification accuracy of approximately 96% on smaller instances and 72% on larger instances, compared with approximately 91% and 55% for NeuroSAT. They also report faster neural inference than running a heuristic solver on the corresponding prediction tasks. For key inference, two derived ANF instances are created by assigning a selected key bit to zero and one. The neural scores are

compared to determine the predicted bit value, improving the reported key-decryption accuracy on two datasets.

The principal limitation is that satisfiability classification is not equivalent to exact solving. A classifier may operate faster because it can return an incorrect answer, whereas a traditional solver must prove satisfiability or unsatisfiability. Key-bit prediction errors also compound as more bits are inferred, limiting reliable full-key recovery. The experiments are restricted to selected plaintext–ciphertext and multivariate-quadratic instances.

For the proposed project, this work demonstrates why cryptographic structure should be preserved. However, replacing the exact solver with CryptoANFNet would weaken correctness. A more appropriate design retains XOR-aware exact constraints and uses the neural model only to propose candidate key assumptions.

Shafran et al. (2024)

Shafran et al. (2024) provide a theoretical investigation of whether machine-learning cryptanalysis is fundamentally restricted compared with conventional attacks. Their research problem concerns how neural networks access training samples. A conventional cryptanalytic algorithm may inspect and manipulate individual plaintext–ciphertext pairs directly, whereas gradient-based learning normally accesses aggregate information through derivatives of a loss function averaged over the dataset. This difference creates the possibility that gradient-based attacks are inherently weaker.

The authors formalise two classes of adversary. A sample-based adversary receives direct access to a complete sample set. A gradient-based adversary receives only the dataset size and accesses the samples through a gradient oracle. The oracle accepts a model, loss function, and parameter vector and returns a gradient averaged over the supplied data. Several simulation definitions are introduced to determine whether the output distribution of a gradient-based adversary approximates that of a sample-based adversary and to quantify the resulting running-time and query overhead.

The theoretical constructions show that a general sample-based adversary can be simulated by a gradient-based adversary. The first construction extracts sample information through a depth-first procedure and reproduces the output distribution exactly. Further constructions support statistical simulation using parallel gradient queries and a procedure closer to ordinary gradient descent. Shafran et al. (2024) argue that the resulting overhead is close to optimal within their formal model. Existing neural attacks against SPECK and learning-with-errors problems are also positioned within the framework.

The result should not be interpreted as evidence that a practical neural network can automatically reproduce every conventional cryptanalytic algorithm. The simulation permits highly flexible models and loss functions designed for theoretical information extraction. These constructions may not resemble efficient or trainable neural architectures. The study does not demonstrate an empirical improvement against a cipher, select a practical architecture, or resolve the data-efficiency and interpretability issues found in experiments.

For the proposed project, the paper supports the theoretical legitimacy of exploring neural assistance. Nevertheless, the contribution must remain empirical and determine whether a compact ResNet supplies useful SPECK subkey rankings after training and inference costs are included.

Zhang et al. (2025)

Zhang et al. (2025) extend neural cryptanalysis beyond differential distinguishers by applying neural networks to integral cryptanalysis of SKINNY. Their research problem is that automated searches for integral properties, including division-property, MILP, and meet-in-the-middle methods, operate over extremely large spaces and may fail to locate optimal combinations of ciphertext bits. The authors investigate whether neural models can discover useful integral behaviour and whether those learned features can be converted into conventional cryptanalytic results.

The method preprocesses sets of ciphertexts using parity-related representations associated with integral properties. Neural networks are trained to identify non-random relationships within these sets. Rather than treating the neural output as the final attack, the authors analyse which linear or nonlinear combinations of ciphertext bits explain the predictions. Boolean-function analysis is then used to translate the learned behaviour into interpretable integral distinguishers. The resulting information is incorporated into an improved automatic meet-in-the-middle search framework.

The study reports that neural models identify integral distinguishers using fewer active input bits than existing automated searches in selected SKINNY configurations. The refined classical search discovers longer key-independent and key-dependent integral distinguishers. A neural-discovered seven-round property is converted into a conventional attack and extended using forward encryption and backward decryption. Zhang et al. (2025) report an improved 16-round key-recovery attack on SKINNY-(n)-(n), together with attacks on 18-round and 20-round variants with larger tweakey configurations.

A major strength of the study is that the learned feature is interpreted and transformed into classical cryptanalysis instead of being accepted as an unexplained classifier output. However, the work concerns integral properties of SKINNY rather than differential ranking in SPECK.

Identifying the underlying combination function still requires extensive enumeration and specialist cryptanalytic analysis. The authors recognise that more efficient feature-search strategies remain necessary.

This paper provides a useful model for the proposed project. The strongest contribution is not that a neural network independently breaks a cipher, but that neural information directs a mathematically verifiable procedure. The proposed neural–SAT system follows the same principle by using neural ranking as guidance while retaining exact key recovery and verification.

“Neural-Guided Enumerative SAT Framework” (2026)

The anonymous manuscript “Neural-Guided Enumerative SAT Framework” (2026) presents the closest prior work to the proposed project because it directly combines neural prediction with SAT-based cryptographic key recovery. The uploaded paper is an anonymous ICLR 2026 submission under review, so its findings should be treated as recent prepublication evidence rather than as an accepted conference result. Its research problem is that conventional SAT solvers become increasingly expensive as cryptographic formula size and key space grow, while end-to-end neural solvers lack the accuracy required for dependable full-key recovery.

Two complementary frameworks are proposed. The solver-centred framework trains a neural model to identify a subset of (k) critical key variables from plaintext–ciphertext information. Candidate assignments to the selected variables are enumerated, and the resulting formulas are passed to a conventional heuristic SAT solver. The network is invoked once near the beginning rather than during every branching decision. The second framework uses ANF encodings and trains a discriminator to classify candidate assignments for a small subset of key variables. The selected assignments are then propagated through simplified equations, allowing further variables to be inferred through unit equations.

The solver-centred framework is evaluated on SAT4CryptoBench and reportedly achieves speed-ups of up to five times over its corresponding solver baseline. The ANF discriminator reportedly reaches approximately 79.5% full-key recovery accuracy on a SIMON-6-32-64 dataset. The study argues that neural-guided partial enumeration can reduce the compounding errors associated with predicting every key bit independently.

Several limitations remain relevant. Enumeration still grows exponentially with the number of selected variables. The results depend on specific benchmark families and cipher configurations and do not establish an attack on a complete modern cipher. The neural experiments were also conducted using A100-class GPU resources. Most importantly, the framework does not centre on identifying when neural guidance is unreliable or when it should be disabled.

This paper removes any defensible claim that neural–SAT cryptanalysis has not been applied to lightweight ciphers. The proposed research must instead differentiate itself through confidence-gated guidance, native XOR-aware incremental assumptions, classical-ranking comparisons, end-to-end cost accounting, and explicit analysis of when neural guidance slows rather than accelerates solving.

Sakagami et al. (2026)

Sakagami et al. (2026) investigate whether large language models can improve neural distinguishers for SPECK32/64. Their research problem arises from the limited performance of conventional convolutional and residual networks at higher encryption rounds. Since LLMs demonstrate broad pattern-recognition and few-shot abilities in natural-language tasks, the authors test whether these capabilities transfer to statistical classification of ciphertext pairs.

The study proposes prompt-based neural distinguishers using several large language models. Ciphertext pairs, hexadecimal representations, and optional XOR differences are supplied through different prompt formats. The experiments examine whether LLMs outperform an existing ResNet baseline, whether modifying the selected input difference or training distribution improves higher-round performance, and how prompt design influences classification. Few-shot examples, ciphertext-only prompts, XOR-enhanced prompts, and chain-of-thought instructions are evaluated. The target remains reduced-round SPECK32/64.

The primary result is negative. Sakagami et al. (2026) find no observable overall improvement over the specialised ResNet baseline. Both the LLM and ResNet approaches lose distinguishing power as the number of rounds increases. The authors also find that selecting different input differences has little impact at higher rounds. Prompt representation does affect performance: explicitly including the XOR result between ciphertext pairs can improve LLM classification relative to less structured prompts. Chain-of-thought prompting does not produce a measurable benefit, which is consistent with the task requiring recognition of bit-level algebraic distributions rather than verbal reasoning.

The findings apply only to the evaluated models, prompts, sample quantities, and SPECK configurations. They do not prove that every future foundation model will be ineffective. Prompt-based evaluation may also be less efficient, more expensive, and less reproducible than training a compact local network. The work primarily evaluates the distinguishing phase and does not present a complete LLM-guided exact key-recovery attack.

For the proposed project, the paper helps reject an unnecessary architecture direction. A compact bit-oriented ResNet is faster, locally trainable, and at least as effective for the selected task. The relevant open problem is whether its weak but measurable statistical signal can reduce the search cost of an exact SAT solver.

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